

Ex. No:6

Use Sleuth Kit to analyze digital evidence

Date:

Aim :

To use The Sleuth Kit (TSK) to analyze a disk image (4Dell Latitude CPi.E01) and examine its partitions, file system, files, deleted files, file metadata and timeline as digital forensic.

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

```
C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users\rakes\Downloads\4Dell Latitude CPi (1).E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot   Start      End          Length    Description
000:  Meta   0000000000  0000000000  0000000001  Primary Table (#0)
001:  -----  0000000000  0000000062  0000000063  Unallocated
002:  000:000  0000000063  0009510479  0009510417  NTFS / exFAT (0x07)
003:  -----  0009510480  0009514259  0000003780  Unallocated

C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

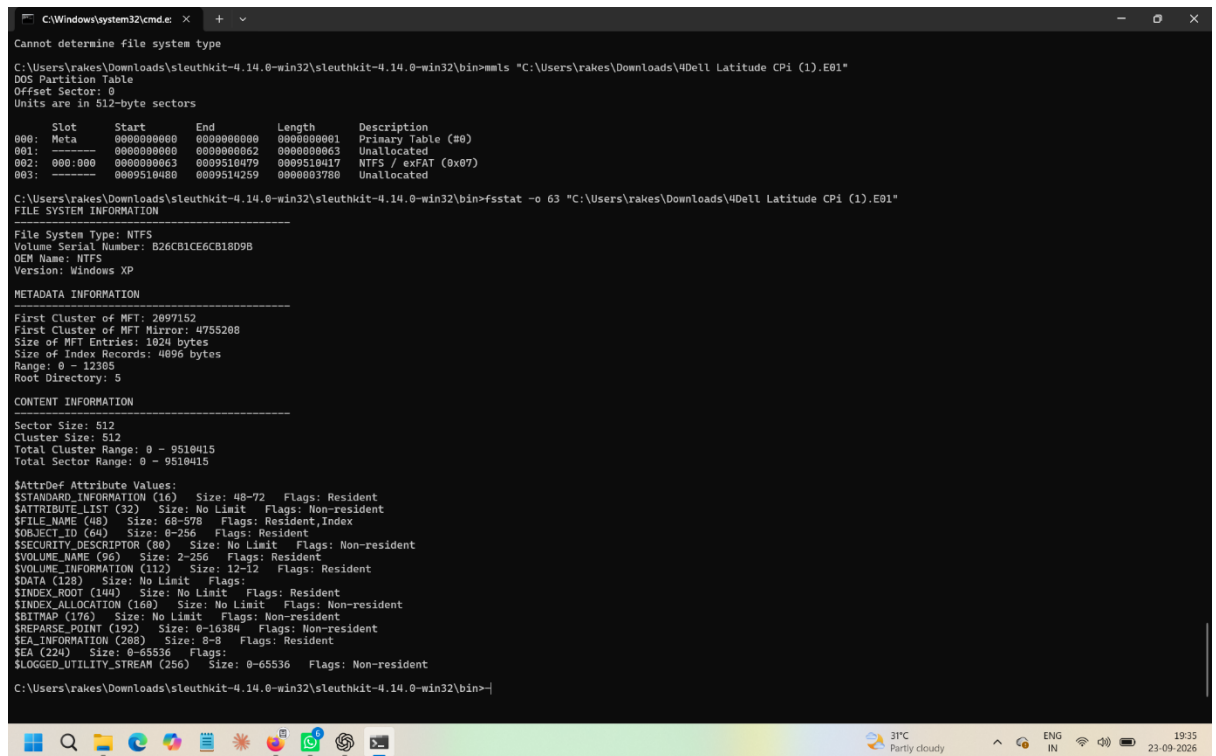
- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.



```
C:\Windows\system32\cmd.exe
Cannot determine file system type

C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users\rakes\Downloads\4Dell Latitude CPl (1).E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

   Slot   Start      End          Length    Description
   ----   -
000:  Meta      000000000000  000000000001  Primary Table (#0)
001:  -----  000000000000  000000000062  Unallocated
002:  000:000  000000000063  00095104179   NTFS / exFAT (0x07)
003:  -----  00095104088  0009514259    Unallocated

C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "C:\Users\rakes\Downloads\4Dell Latitude CPl (1).E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB1809B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755288
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16) Size: 48-72 Flags: Resident
$ATTRIBUTE_LIST (32) Size: No Limit Flags: Non-resident
$FILE_NAME (48) Size: 68-978 Flags: Resident, Index
$OBJECT_ID (64) Size: 0-256 Flags: Resident
$SECURITY_DESCRIPTOR (80) Size: No Limit Flags: Non-resident
$VOLUME_NAME (96) Size: 2-256 Flags: Resident
$VOLUME_INFORMATION (112) Size: 12-12 Flags: Resident
$DATA (128) Size: No Limit Flags:
$INDEX_ROOT (144) Size: No Limit Flags: Resident
$INDEX_ALLOCATION (160) Size: No Limit Flags: Non-resident
$BITMAP (176) Size: No Limit Flags: Non-resident
$REPARSE_POINT (192) Size: 0-16384 Flags: Non-resident
$EA_INFORMATION (208) Size: 0-8 Flags: Resident
$EA (224) Size: 0-65536 Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536 Flags: Non-resident

C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>|
```

- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

fsstat [image file] > filesystem_info.txt

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

mmls [image file] > partitions.txt

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

fls -r [image file] > file_list.txt

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

icat [image file] [inode number] > [output file]

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

```
C:\Windows\system32\cmd.exe
++ r/r 4034-128-4: smaller.fon
++ r/r 4062-128-4: smallet.fon
++ r/r 3896-128-4: smallf.fon
++ r/r 3932-128-4: smallfe.fon
++ r/r 3996-128-4: smallfg.fon
++ r/r 4035-128-4: smallfr.fon
++ r/r 4063-128-4: smallft.fon
++ r/r 4036-128-4: sserifer.fon
++ r/r 4064-128-4: sserifet.fon
++ r/r 3998-128-4: sseriffg.fon
++ r/r 4037-128-4: sseriffrr.fon
++ r/r 4065-128-4: sseriffrr.fon
++ r/r 3923-128-4: cga40852.fon
++ r/r 3470-128-3: ega40850.fon
++ r/r 3425-128-3: mvboli.ttf
++ r/r 3412-128-3: smalle.fon
++ r/r 3455-128-3: timesbd.ttf
++ r/r 4000-128-4: vga869.fon
++ r/r 3965-128-4: vgaf1257.fon
++ r/r 3403-128-3: vgafix.fon
++ r/r 3936-128-4: vgafixe.fon
++ r/r 4040-128-4: vgafixr.fon
++ r/r 4067-128-4: vgafixt.fon
++ r/r 4041-128-4: vgasysr.fon
++ r/r 4068-128-4: vgasyst.fon
+ r/r 5028-128-4: Gone Fishing.bmp
+ r/r 5029-128-4: Greenstone.bmp
+ d/d 9846-144-6: Installer
++ r/r 10179-128-3: ac704.msi
++ r/r 9860-128-3: d2a9f.msi
++ d/d 9861-144-1: {350C97B0-3D7C-4EE8-BAA9-00BCB3D54227}
+++ r/r 9862-128-3: places.exe
++ d/d 10180-144-6: {6C31E111-96BB-4ADC-9C81-E6D3EEDDD8D3}
+++ r/r 10183-128-4: ARPPRODUCTICON.exe
+++ r/r 10181-128-3: htmlgen.exe
+++ r/r 10182-128-4: PowerCalc.exe
+++ r/r 10185-128-3: Readme
+++ r/r 10184-128-4: Timershot.exe
+++ r/r 10186-128-3: TweakUI.exe
+ r/r 11772-128-3: iun6002.exe
+ r/r 11734-128-4: Look@LAN Setup Log.txt
+ r/r 3651-128-3: explorer.scf
+ r/r 3653-128-3: msdfmap.ini
+ r/r 339-128-3: REGLOCS.OLD
+ r/r 3680-128-1: setuperr.log
+ r/r 5464-128-1: vb.ini
+ r/r 3901-128-4: regopt.log
+ r/r 431-128-3: SchedLgU.Txt
+ r/- * 0: ~GLC0000.TMP
+ r/- * 0: ~GLH0000.TMP
+ r/- * 0: ~GLH0001.TMP
V/V 12305: $OrphanFiles
r/r 343-128-1: hiberfil.sys
r/r 228-128-1: pagefile.sys
C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

CSS

istat [image file] [inode number] > metadata_info.txt

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

CSS

fls -m / -r [image file] > body.txt

- Then create the timeline:

CSS

mactime -b body.txt > timeline.txt

```
+ r/r 4747-128-1: wiadebug.log
+ r/r 4745-128-1: wiaservc.log
+ r/r 3647-128-3: win.ini
+ r/r 6611-128-1: Windows Update.log
+ r/r 6323-128-5: WindowsShell.Manifest
+ r/r 3658-128-3: winhlp.exe
+ r/r 3659-128-3: winhlp32.exe
+ r/r 5716-128-4: winnt.bmp
+ r/r 5717-128-4: winnt256.bmp
+ d/d 3634-144-6: WinSxS
++ d/d 3636-144-4: InstallTemp
++ d/d 3635-144-7: Manifests
+++ r/r 3695-128-5: x86_Microsoft.Tools.VisualStudio.CPlusPlus.Runtime-Libraries_6595b64144ccf1df_6.0.0.0_x-ww_ff9986d7.cat
+++ r/r 3696-128-5: x86_Microsoft.Tools.VisualStudio.CPlusPlus.Runtime-Libraries_6595b64144ccf1df_6.0.0.0_x-ww_ff9986d7.Manifest
+++ r/r 3699-128-5: x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.0.0_x-ww_1382d70a.cat
+++ r/r 3700-128-5: x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.0.0_x-ww_1382d70a.Manifest
+++ r/r 3704-128-5: x86_Microsoft.Windows.CPlusPlusRuntime_6595b64144ccf1df_7.0.0.0_x-ww_2726e76a.cat
+++ r/r 3705-128-5: x86_Microsoft.Windows.CPlusPlusRuntime_6595b64144ccf1df_7.0.0.0_x-ww_2726e76a.Manifest
+++ r/r 3686-128-5: x86_Microsoft.Windows.GdiPlus_6595b64144ccf1df_1.0.0.0_x-ww_8d353f13.cat
+++ r/r 3687-128-5: x86_Microsoft.Windows.GdiPlus_6595b64144ccf1df_1.0.0.0_x-ww_8d353f13.Manifest
+++ r/r 3688-128-5: x86_Microsoft.Windows.SystemCompatible_6595b64144ccf1df_5.1.0.0_x-ww_fc342b0b.cat
+++ r/r 3689-128-5: x86_Microsoft.Windows.SystemCompatible_6595b64144ccf1df_5.1.0.0_x-ww_fc342b0b.Manifest
++ d/d 3690-144-10: x86_Microsoft.Tools.VisualStudio.CPlusPlus.Runtime-Libraries_6595b64144ccf1df_6.0.0.0_x-ww_ff9986d7
+++ r/r 3693-128-4: atl.dll
+++ r/r 3692-128-4: mfc42.dll
+++ r/r 3691-128-4: mfc42u.dll
+++ r/r 3694-128-4: msvcpr60.dll
++ d/d 3697-144-1: x86_Microsoft.Windows.Common-Controls_6595b64144ccf1df_6.0.0.0_x-ww_1382d70a
+++ r/r 3698-128-4: comctl32.dll
++ d/d 3701-144-7: x86_Microsoft.Windows.CPlusPlusRuntime_6595b64144ccf1df_7.0.0.0_x-ww_2726e76a
+++ r/r 3703-128-4: msvcirt.dll
+++ r/r 3702-128-4: msvert.dll
++ d/d 3684-144-1: x86_Microsoft.Windows.GdiPlus_6595b64144ccf1df_1.0.0.0_x-ww_8d353f13
+++ r/r 3685-128-4: GdiPlus.dll
+ r/r 7284-128-4: WMSysPrx.prx
+ r/r 5034-128-4: Zapotec.bmp
+ r/r 3650-128-3: _default.pif
+ r/r 4098-128-4: FaxSetup.log
+ r/r 5027-128-4: FeatherTexture.bmp
+ d/d 3400-144-6: Fonts
++ r/r 3456-128-3: timesbi.ttf
++ r/r 3457-128-3: timesi.ttf
++ r/r 3430-128-3: trebuch.ttf
++ r/r 3431-128-3: trebuchd.ttf
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

```
C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>dir "C:\Users\rakes\Downloads\Gone_Fishing_recovered.bmp"
Volume in drive C is Windows-SSD
Volume Serial Number is DC08-8AC2

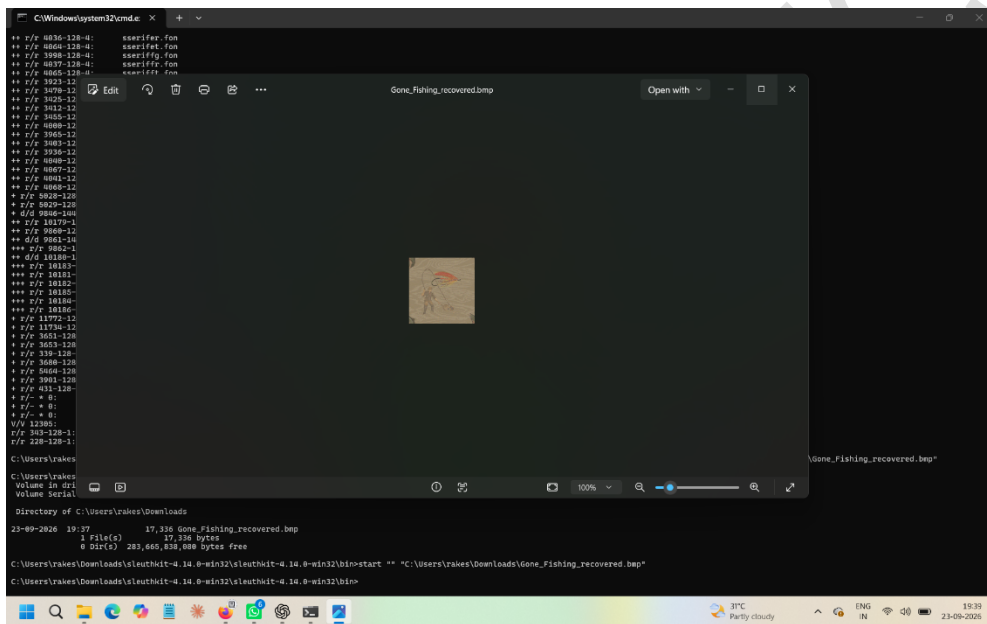
Directory of C:\Users\rakes\Downloads

23-09-2026  19:37                17,336 Gone_Fishing_recovered.bmp
             1 File(s)                17,336 bytes
             0 Dir(s) 283,665,838,080 bytes free

C:\Users\rakes\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.



Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

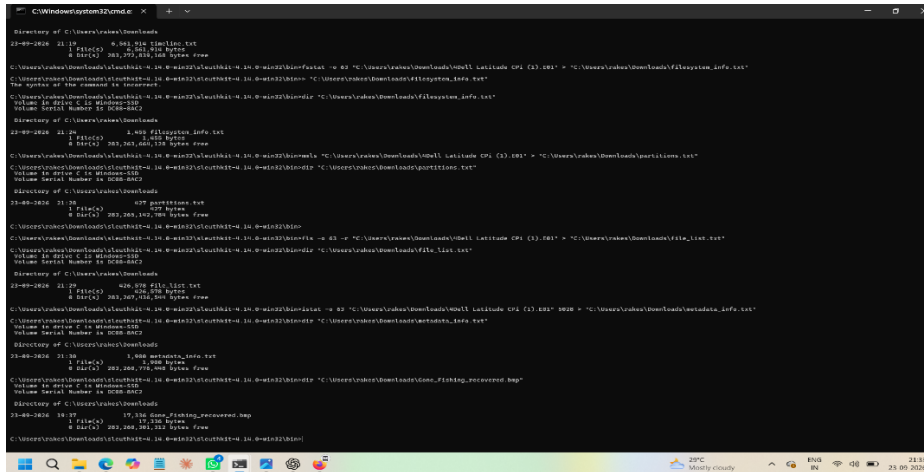
```
File Edit View H1 125 B I S C O P A D
-- Strawberry Perl (64-bit) 5.42.0.1-64bit README ---
What is Strawberry Perl?
* Perl is a programming language suitable for writing simple scripts as well
  as complex applications. See https://perldoc.perl.org/perlintro.html
* Strawberry Perl is a perl environment for Microsoft Windows containing all
  you need to run and develop perl applications. It is designed to be as close
  as possible to perl environment on Unix systems. See https://strawberryperl.com/
* If you are completely new to perl consider visiting https://learn.perl.org/
  Installation instructions (ZIP distribution only, not .exe installer)
  https://strawberryperl.com/zip-distribution
* If installing this version from a .zip file, you MUST extract it to a
  directory that does not have spaces in it, e.g. c:\perl
  and then run some commands and manually set some environment variables:
  c:\perl\release\pl.bat ... this is REQUIRED!
  c:\perl\update_per.bat ... this is OPTIONAL
  You can specify "--noupdate" after update_per.pl.bat to install Strawberry
  Perl's environment variables for the current user only.
  If having a fixed installation path does not suit you, try "Strawberry Perl
  Portable Edition" from https://strawberryperl.com/releases.html
How to use Strawberry Perl?
* In the command prompt window you can:
  1. run any perl script by launching
     c:\> perl c:\path\to\script.pl
  2. install additional perl modules (libraries) from https://www.metacpan.org/ by
     c:\> cpan Module::Name
  3. run other tools included in Strawberry Perl like: perlbug, gcc, make ...
* You'll need a text editor to create perl scripts. One is not included with
  Strawberry Perl, a few custom ones (which use the installed perl compiler)
  "nano" (from https://www.nano-editor.org/), "vim" (from https://www.vim.org/),
  "notepad-plus-plus" (from https://notepad-plus-plus.org/) which both include syntax highlighting
  for perl scripts. You can even use Notepad, if you wish.
```

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.



By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Rubric:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result: The Sleuth Kit was successfully used to analyze the forensic disk image. The file system, partitions, files, and file metadata were examined, a deleted file was recovered, and a forensic timeline was generated successfully. Thus, the experiment demonstrated the use of Sleuth Kit for digital evidence analysis and investigation.